

Generic Robotics Simulators

- mujoco
- O3DE for Robotics
- Webots
- Gazebo
- Newton (Physics)
- OpenRAVE
- Unity Robotics Hub
- NVIDIA Isaac Sim
- Ros2 For Unity
- ARGoS
- ROS1 Intergration for Unreal 4
- CoppeliaSim core library
- Robwork
- SimSpark
- CoppeliaSim
- RoboDK

Robotic simulators in 2D

- IR-SIM
- pyrobosim
- mvsim
- Flatland

Aerial Robotics Simulators

- Colosseum
- Cosys-AirSim
- PyFlyt
- Gym Pybullet Drones
- RotorS
- Flightmare
- Pegasus Simulator
- Project AirSim
- FlightGoggles
- rotorpy
- Aerial Gym Simulator

- jMAVSim
- Crazyflow
- Gymfc
- CrazyS
- Isaac Drone Racer
- Fly Craft
- Flightgear
- FlightForge
- Ardupilot_Multiagent_Simulation
- Itomori

Maritime Robotics Simulators

- Virtual RobotX
- DAVE
- UNav-Sim
- Stonefish
- Ocean Sim
- Moos-ivp
- HoloOcean
- MARUSimulator

Automotive Simulators

- Carla
- ESMINI
- AWSim

Space Robotics Simulators

- Basilisk
- Astrobee
- BSK-RL
- OmiLRS
- Mars Explorer
- LunarSim

AI training Simulators

- Gymnasium
- Genesis

- NVIDIA Isaac Sim Isaac Lab
- Metaworld
- AI2-thor
- Gymnasium Robotics
- jaxsim
- IGibson
- Habitat Sim
- Jiminy
- ManiSkill
- Sapien
- MuJoCo playground
- RoboVerse
- MetaDrive
- K-Sim
- mjlab
- Holodeck
- LocoMuJoCo
- Gibson
- Safe Control Gym
- PyRep
- robo-gym
- ProcTHOR
- RoboCasa
- Deepdrive
- racecar_gym
- OmnisaacGymEnvs
- PyBullet Gym
- UAlbot
- Arnold
- IsaacLab-HARL

Other Domain Specific Simulators

- Panda-Gym
- AutoDRIVE Simulator
- DexterousHands
- Graspit!
- gym-softrobot
- Strech MuJoCo

- Stewart Platform Simulator

Physics Engines

- Drake
- Pinocchio
- Project CHRONO
- Bullet Physics SDK
- BRAX
- JSBSim
- DART
- Simbody
- MuJoCo Wrap
- PhysX 5
- TPE (part of gz-physics)
- RaiSim
- ODE

Others

- HighwayEnv

<https://github.com/jslee02/awesome-robotics-libraries>

Simulators

- AI2-THOR
- AirSim
- ARGoS
- ARTE
- AVIS Engine
- CARLA
- Coppeliasim
- Gazebo
- Graspl
- Habitat-Sim
- Hexapod Robot Simulator
- Ignition Gazebo
- Isaac Sim

- ManiSkill
- MORSE
- Neurorobotics Platform
- PyBullet
- PyBullet_Industrial
- SAPIEN
- Simbad
- Unity
- Webots
- Actin Simulation
- Kineo
- RobotDK
- RobotStudio
- Robot Virtual Worlds
- Visual Components
- AWS RoboMaker

Libraries - Dynamics Simulation

- ARCSim
- Bullet
- CHRONO::ENGINE
- DART
- Drake
- Flex
- FROST
- IBDS
- idyntree
- KDL
- kindr
- Klampt
- MBDyn
- MBSim
- MBSlib
- Moby
- MuJoCo
- mvsim
- Newton Dynamics
- nphysics

- ODE
- OpenRAVE
- pinocchio
- PositionBasedDynamics
- PhysX
- PyDy
- RBDL
- RBDyn
- RaiSim
- ReactPhysics3d
- RigidBodyDynamics.jl
- Rigs of Rods
- Robopy
- Robotics Library
- RobWork
- siconos
- Simbody
- SOFA
- Tiny Differentiable Simulator
- trep
- qu3e

Libraries - Motion Planning and Control

- AIKIDO
- Bioptim
- CuiKSuite
- cuRobo
- Control Toolbox
- Crocoddyl
- Fields2Cover
- GPMP2
- HPP
- MoveIt!
- OMPL
- OCS2
- pymanoid
- Ruckig
- The Kautham Project

- TOPP-RA
- Ungar

<https://github.com/ahundt/awesome-robotics>

- CoppeliaSim
- Microsoft Airsim
- Bullet Physics SDK

<https://github.com/kiloreux/awesome-robotics>

- Gazebo Robot Simulator
- RobWork
- MRPT
- Robotics Library
- Simbad
- Morse
- V-REP
- Webots
- Drake
- Neurorobotics Platform
- Open AI's Roboschool
- RBDL
- Unity Robotics Hub

<https://github.com/AGV-IIT-KGP/awesome-motion-planning>

- OMPL
- SBPL
- SMPL

<https://github.com/Ly0n/awesome-robotic-tooling>

Simulation

- AI2-THOR
- Drake
- Webots
- lgsv
- carla
- uuv_simulator
- AirSim
- gym-gazebo
- gym-pybullet-drones
- safe-control-gym
- highway-env
- VREP Interface
- sumo
- open-simulation-interface
- ESIM
- Menge
- pedsim_ros
- morse
- fetch_gazebo
- rotors_simulator
- flow
- gnss-ins-sim
- Ignition Robotics
- gazebo_ros_motors
- sim_vehicle_dynamics
- gym-carla
- simbody
- gazebo_models
- pilot
- flightmare
- champ
- rex-gym
- Trick
- usv_sim_lsa
- 42
- AutoCore simulation
- fields-ignition
- Unity-Robotics-Hub

- BlueSky
- Cloe
- Dynamic_logistics_Warehouse
- OpenCDA

Planning and Control

- rrt
- HypridAStarTrailer
- path_planner
- commonroad
- steering_functions
- moveit
- flexible-collision-library
- aikido
- casADi
- CrowdNav
- ompl
- openrave
- teb_local_planner
- pinocchio
- OpEn
- global_racetrajectory_optimization
- toppra
- tinyspline
- EGO-Planner
- pykep
- am_traj
- GraphBasedLocalTrajectoryPlanner
- se2_navigation
- Ruckig

<https://github.com/fkromer/awesome-ros2>

- ros2
- rviz
- urdfdom

<https://github.com/lukicdarkoo/awesome-webots>

- Webots
- Eurobot Platform

<https://github.com/mjyc/awesome-robotics-projects>

Simulators

- AI Habitat
- CARLA
- Flightmare
- Gazebo
- Gibson
- LIBERO
- MetaDrive
- MORSE
- NVIDIA Isaac Sim
- OmniGibson
- Open 3D Engine
- Project dave
- SimplerEnv
- Unity Robotics Hub
- Webots
- ZeroSimROSUnity

https://en.wikipedia.org/wiki/Robotics_simulator

Simulators

- Gazebo
- RoboDK
- SimSpark
- Webots
- OpenRAVE
- CoppeliaSim

Žlajpah (2008) – “Simulation in Robotics”

- MATLAB / Simulink
- SimMechanics
- Dymola / Modelica
- 20-sim
- Robotics Toolbox
- SD/FAST
- Graspl!
- Open Dynamics Engine
- Newton Game Dynamics
- V-REP (CoppeliaSim)
- ABB RobotStudio
- RobotWorks
- MotoSim
- ROBCAD / Workcell
- KiKS

Iñigo-Blasco et al. (2012) – “A Survey of Development Frameworks for Robotics”

- ROS
- Orocos
- OpenRTM-aist
- Gazebo
- YARP
- MRDS
- URBI
- CLARAty
- CARMEN
- GenoM

Körber et al. (2021) – “Comparing Popular Simulation Environments in the Scope of Robotics and Reinforcement Learning”

- Gazebo
- MuJoCo
- PyBullet
- Webots
- Bullet
- DART
- Simbody
- PhysX

Collins et al. (2021) – “A Review of Physics Simulators for Robotic Applications”

- Gazebo
- CoppeliaSim (V-REP)
- Webots
- MuJoCo
- PyBullet
- RaiSim
- CARLA
- UUV Simulator
- UWSim
- SOFA
- CHAI3D
- AMBF
- NVIDIA Isaac / Flex
- Project Chrono
- Blender

Brugali et al. (2008) – “A Survey of Development Frameworks for Robotics”

- ROS

- Orocos
- OpenRTM-aist
- Player / Stage
- Gazebo
- URBI
- YARP
- GenoM
- MRDS
- CLARAty

Kargar et al. (2024) – “Emerging Trends in Realistic Robotic Simulations”

- Gazebo
- Webots
- CoppeliaSim (V-REP)
- MuJoCo
- PyBullet
- Isaac Sim
- Unity
- MORSE
- ARGoS
- PhysX
- Bullet
- ODE
- DART

Farley et al. (2022) – “How to Pick a Mobile Robot Simulator”

- Gazebo
- Webots
- MORSE
- CoppeliaSim (V-REP)
- ROS
- Bullet
- ODE

- DART
- Simbody
- Vortex
- Newton

Harris & Conrad (2011) – “Survey of Popular Robotics Simulators, Frameworks and Toolkits”

- ROS
- Gazebo
- OpenRAVE
- OpenRTM-aist
- MRPT
- SimRobot
- USARSim
- MRDS
- Moby

Staranowicz & Mariottini (2011) – “A Survey and Comparison of Commercial and Open-Source Robotic Simulation Software”

- Webots
- Easy-Rob
- MATLAB / Simulink + Robotics Toolbox
- Gazebo
- USARSim
- CARMEN
- MRDS
- SARGE

Reckhaus et al. (2010) – “An Overview about Simulation and Emulation in Robotics (SIMPAR Workshops)”

- Graspl!
- OpenRAVE
- Player / Stage / Gazebo
- MATLAB / Simulink
- Spice

Harris & Conrad (2011) – “Survey of Popular Robotics Simulators, Frameworks, and Toolkits”

- Player / Stage / Gazebo
- USARSim
- SARGE
- ROS
- UberSim
- EyeSim
- SubSim
- OpenRAVE
- OpenRTM-aist
- MRPT
- Ipzrobots / ode_robots
- SimRobot
- Moby
- MRDS
- Marilou
- Webots
- robotSim Pro
- robotBuilder

gpt

- MoveIt / MoveIt 2
- OMPL

- Drake
- Gazebo
- Webots
- CoppeliaSim
- MuJoCo
- Bullet / PyBullet
- DART
- ODE
- NVIDIA Isaac Sim
- PhysX SDK
- Habitat-Sim
- iGibson / OmniGibson
- ManiSkill
- SAPIEN
- MuJoCo
- Tesseract
- TrajOpt / trajopt_ros
- OpenRAVE
- Pinocchio
- RBDL
- iDynTree
- Orocos KDL
- ros2_control (ros_control)
- Navigation2 (Nav2)
- FCL (Flexible Collision Library)
- SOFA
- CrazySwarm / CrazySwarm2
- Polymetis
- acados
- CasADi
- do-mpc

gemini

- ROS (Robot Operating System)
- ROS 2
- Drake
- YARP (Yet Another Robot Platform)

- Orocos (Open Robot Control Software)
- MoveIt
- OMPL (Open Motion Planning Library)
- SBPL (Search-Based Planning Library)
- TrajOpt
- CHOMP (Covariant Hamiltonian Optimization)
- STOMP (Stochastic Trajectory Optimization)
- KineoWorks
- Robotics Toolbox for MATLAB
- MuJoCo (Multi-Joint dynamics with Contact)
- Bullet Physics (and PyBullet)
- ODE (Open Dynamics Engine)
- PhysX
- Pinocchio
- KDL (Kinematics and Dynamics Library)
- Crocoddyl (Contact RObot COntrol)
- Gazebo
- NVIDIA Isaac Sim
- Webots
- CoppeliaSim (formerly V-REP)
- AirSim
- RoboDK
- RobotStudio
- Delmia